COMPUTER GRAPHICS

ASSIGNMENT #1

WIRE FRAME MODELER

INTRODUCTION

In this assignment you will develop the first stage of a modeling software. Your program will render models in wire-frame (edges only) . The emphasis in this assignment is the correct application of transformation and the design of a GUI.

REQUIREMENTS

You are given a code skeleton for your program. You are required to use the skeleton.

The skeleton has "TODO(student)" comments in places for you to fill in, so it's easy to search and see what's missing.

About the skeleton structure:

- math.h: A linear algebra library that implements vectors and matrices. Implement all the missing functions marked with "TODO".
- Models:
    - model.h: Implement loading model geometry from an obj file in LoadModelFromObjFile. If the model has vertex normals and texture coordinates, load those as well. You can use `stb_ds.h` for easy dynamic arrays.
    - primitives.h: create some primitives (at least a cube and a sphere) that are stored as C data embedded in code.
- scene.h: The scene is composed from models and cameras. Add to the Camera struct what information you need to transform a camera, as well as creating orthographic and perspective projections.
- renderer.h: Implement rendering the scene from the active camera into wireframes using Bresenham's line drawing algorithm. Add an support for drawing the model's bounding box (compued in model frame), it's face normals, and it's vertex normals (if those are available). Add support for drawing a widget (like a plus) for other cameras visible in the scene. The distance between two projected points should stay constant regardless of the window size.
- interface.h: Add support for editing the position and orientation values for the cameras and the models, for adding or deleting models and cameras to the scene, for enabling bounding box and normal to be drawn, and for editing the camera parameters and projection type. Add support

for moving the active model or camera with your mouse or keyboard (or both), both in world coordinates and relative to the active camera. Add support for focusing the active camera on a model.

SUBMISSION

You have to submit your code electronically on gr.cs.technion.ac.il. You should submit the entire project, in a zip file, excluding the debug and release folders. You should not have any files larger than 1MB.

You have to schedule a presentation with the TA so the homework you uploaded can be presented. Presentations will last 15-20 minutes, during which you will show the TA your work and answer questions.

FINAL NOTES

- Using AI Bots (e.g., ChatGPT):
    - You are allowed to use AI bots in this course only in certain limited and defined circumstances. You must disclose that you did so, and for what purpose.
        - Allowed is:
            - Using a chatbot (e.g., ChatGPT) to help you compile, link, research an API call, or find an answer to a non CG related question. You should specify which chatbot, and the general purpose (e.g. "help with compilation", or similar).
            - Using an AI-collaborative coding environment (e.g., Co-Pilot, Codex, or Claude Code) to write non CG related code (e.g., input/output, string operations, etc.). This is allowed. You should specify which environment, and the general purpose (e.g. "writing the input/output functions").
        - All other uses are forbidden. Examples include:
            - Using an AI-collaborative coding environment (e.g., Co-Pilot, Codex, or Claude Code) to write CG related code (e.g., parts of the rasterizer, shaders, z-buffer, etc.). Any algorithm that we teach in the lecture or the tutorial is CG related. THIS IS NOT ALLOWED.
            - Using an AI-collaborative coding environment (e.g., Co-Pilot, Codex, or Claude Code) to debug your CG-related code, or help you find and fix mistakes in your CG-related code. THIS IS NOT ALLOWED.
    - If in doubt, ask.
    - You are responsible for the code you submit, and you should be able to explain it.

- This is not MATAM – there is no automatic checker. This means that all the features that are to be implemented should be intuitive to the developers with plenty of room for personal

interpretation. It also means that the features that you implement should behave quite differently compared two different works. Copying of any kind is forbidden!

- DO NOT USE any external code without permission. If you have any doubt, please ask.
- You are very much encouraged to experiment with your program and add more features to it. Previous experiences show this assignment can be addicting!
- You have two weeks to complete this exercise, this time is more than enough, but you are strongly encouraged to start working on it right away.
  We know that this assignment could be intimidating, and possibly the largest assignment you will encounter during your studies. Do not hesitate to ask us for help!